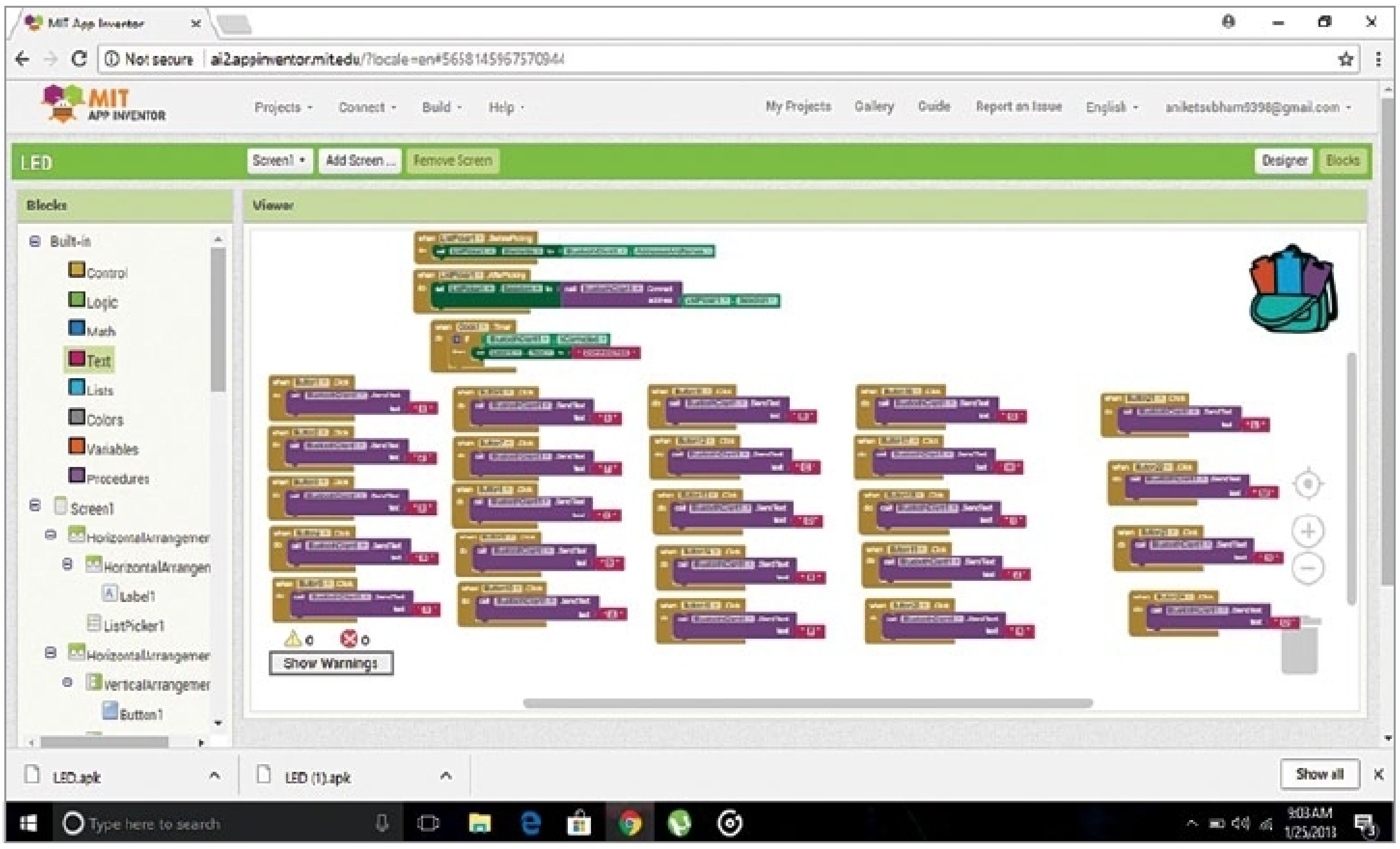

Fig. 3: Screenshot of Designer section in MIT App Inventor

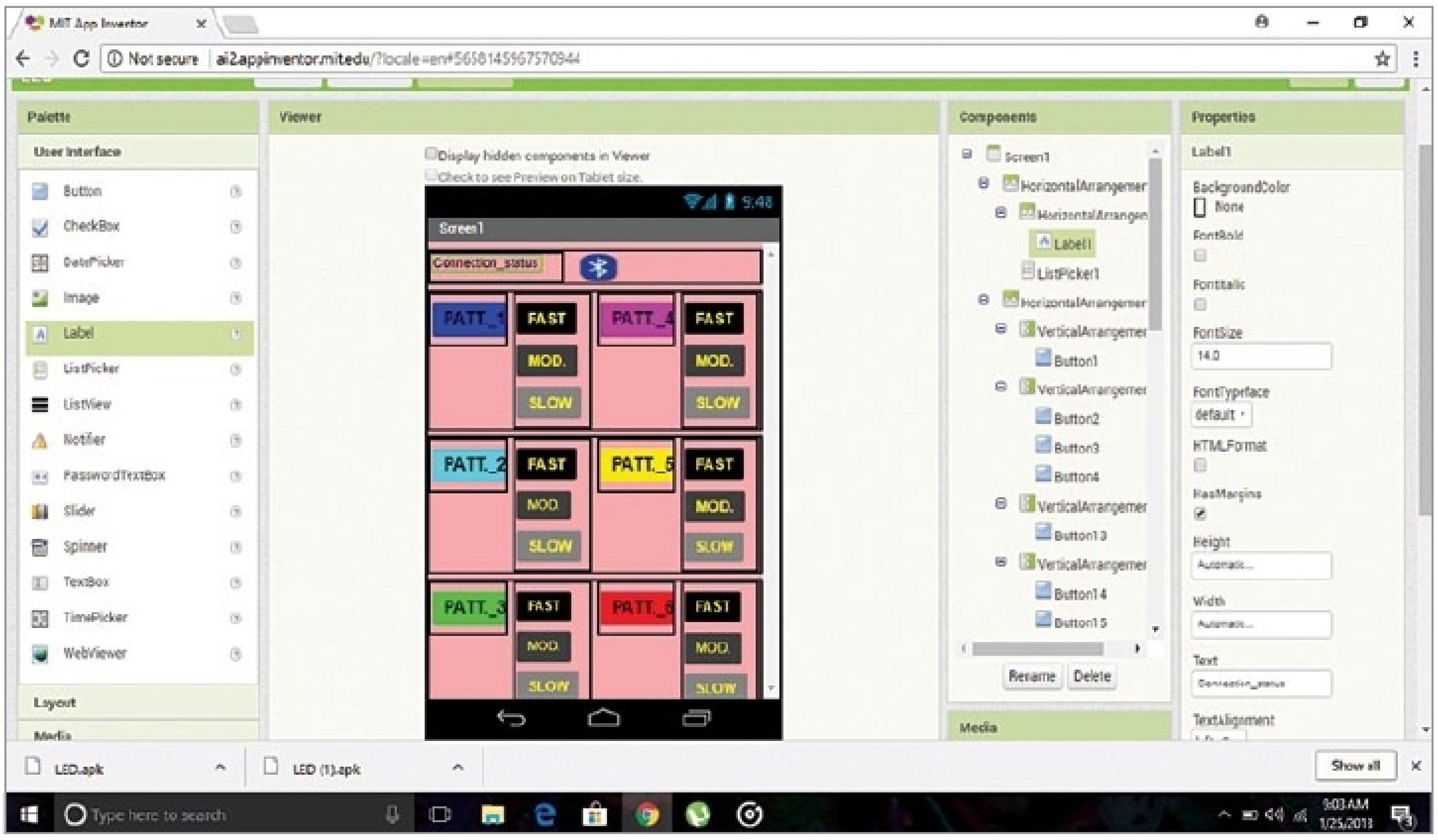

Fig. 4: Screenshot of Blocks section in MIT App Inventor

frequency hopping (AFH) feature.

Android smartphone. The controlling app is based on Android, and has been designed using MIT App Inventor, which is an intuitive, visual programming environment that allows the user to build fully-functional apps for smartphones and tablet PCs. The project has been tested on Redmi 3S smartphone.

Software

For execution of the project, connect Arduino Uno board to the laptop/PC using a USB cable.

Open the source code (LEDchaser_app.ino), and select proper board and COM port under Tools menu. Compile and upload the source code to Arduino Uno.

Serial.println() function helps control the LEDs from Tx pin of Bluetooth to Arduino. Now, make connections as per the circuit diagram (Fig. 2).

Download and install LED.apk on the smartphone. After successful installation, turn on Bluetooth of the smartphone and connect it with HC-05 module; default password for HC-05 is 1234 or 0000.

After successful pairing with Bluetooth and the smartphone, open the app in the smartphone. Now, press the pushbutton with Bluetooth logo. From paired devices list, choose HC-05 device. After successful connection, 'Connected' will be displayed on the app in the smartphone.

You can control the glowing patterns of LEDs in three modes, namely, fast, moderate and slow. Android app can now control the LEDs with six different patterns when you tap on the corresponding button.

MIT App Inventor design

Some important development stages of app development, that is, screenshots of Designer and Blocks section as designed using MIT App Inventor are shown in Figs 3 and 4, respectively.

LED.apk is generated from MIT App Inventor after completion of above stages. **EFY**

EFY Note

The source code of this project is available for free download at *source.efymag.com*

Aniket Subham has a keen interest in designing MCU-based embedded systems

Shibendu Mahata is MTech (gold medallist) in instrumentation and electronics engineering from Jadavpur University. He has keen interest in MCU-based real-time embedded signal processing and process control systems